Two-Track DCA$_{xy}$ Comparison

FastSim closest-point residual vs. cross-product projection

Definitions

For track PCA positions $\mathbf{r}_i$, $\mathbf{r}_j$ and momentum directions $\mathbf{u} = \mathbf{p}_i/|\mathbf{p}_i|$, $\mathbf{v} = \mathbf{p}_j/|\mathbf{p}_j|$, define

$$\mathbf{n} = \frac{\mathbf{u}\times\mathbf{v}}{|\mathbf{u}\times\mathbf{v}|}, \qquad \Delta\mathbf{r} = \mathbf{r}_j - \mathbf{r}_i.$$

The cross-product method discussed here is

$$D_{xy}^{\mathrm{cross}} = \mathbf{n}\cdot(\Delta x, \Delta y, 0).$$

This is a projection of the transverse PCA separation onto the normal vector of the two track directions.

The current FastSim method first solves the two closest points $\mathbf{c}_i$ and $\mathbf{c}_j$ on the full 3D track lines, then computes

$$\boldsymbol{\rho} = \mathbf{c}_i - \mathbf{c}_j, \qquad D_{xy}^{\mathrm{FS}} = s\sqrt{\rho_x^2 + \rho_y^2},$$

where the sign s is set from

$$s = \mathrm{sign}\left[\left((\mathbf{p}_{T,i} + \mathbf{p}_{T,j}) \times (\mathbf{r}_i - \mathbf{r}_j)\right)_z\right].$$

Main Result

These are not the same observable. The FastSim value is the transverse magnitude of the closest-point residual. The cross-product value is a normal-vector projection after forcing the PCA separation's z component to zero. They can have similar overall scale, but they are not related by a single event-by-event scale factor.

Signed values should not be compared directly without first harmonizing the sign convention. The table below compares absolute magnitudes.

Numerical Check

The following check used the default FastSim configuration, seed 20260509, 20,000 generated events, and 12,919 selected Si-Si pairs. All distances are in cm.

Quantity	Median	68%	90%	95%	99%	Max
$\lvert D_{xy}^{\mathrm{FS}}\rvert$	0.00500	0.00819	0.01683	0.02260	0.03636	0.08340
$\lvert D_{xy}^{\mathrm{cross}}\rvert$	0.00419	0.00710	0.01578	0.02174	0.03611	0.08151
$\big\lvert\lvert D_{xy}^{\mathrm{cross}}\rvert - \lvert D_{xy}^{\mathrm{FS}}\rvert\big\rvert$	0.00240	0.00400	0.00844	0.01159	0.01925	0.03683

The ratio $|D_{xy}^{\mathrm{cross}}|/|D_{xy}^{\mathrm{FS}}|$ had median 0.91. Its 5–95% interval was 0.087–6.64, with a 99% value of 32.7. The large upper tail appears when the FastSim denominator is close to zero.

Contrast: DCA vs. DCAsP

In the related FastSim check, the comparison was between DCA and DCAsP. That is a different situation: both observables use the same closest-point residual $\boldsymbol{\rho}$, and only the sign convention changes. Therefore their absolute magnitudes are identical event-by-event,

$$|D_{xy}^{\mathrm{DCA}}| = |D_{xy}^{\mathrm{DCAsP}}|, \qquad |D_z^{\mathrm{DCA}}| = |D_z^{\mathrm{DCAsP}}|.$$

A fresh 20,000-event check with the current code gave 55,629 selected pairs, zero maximum absolute-magnitude difference in both XY and Z, identical RMS values, and sign agreement of 0.499. Thus the DCA/DCAsP resolution plots are expected to be numerically very close; the cross-product projection above is not in that sign-only category. The projection and FastSim definitions may have similar width scales, but their pair-by-pair differences can reach a few 10^{-2} cm in the default toy sample.